

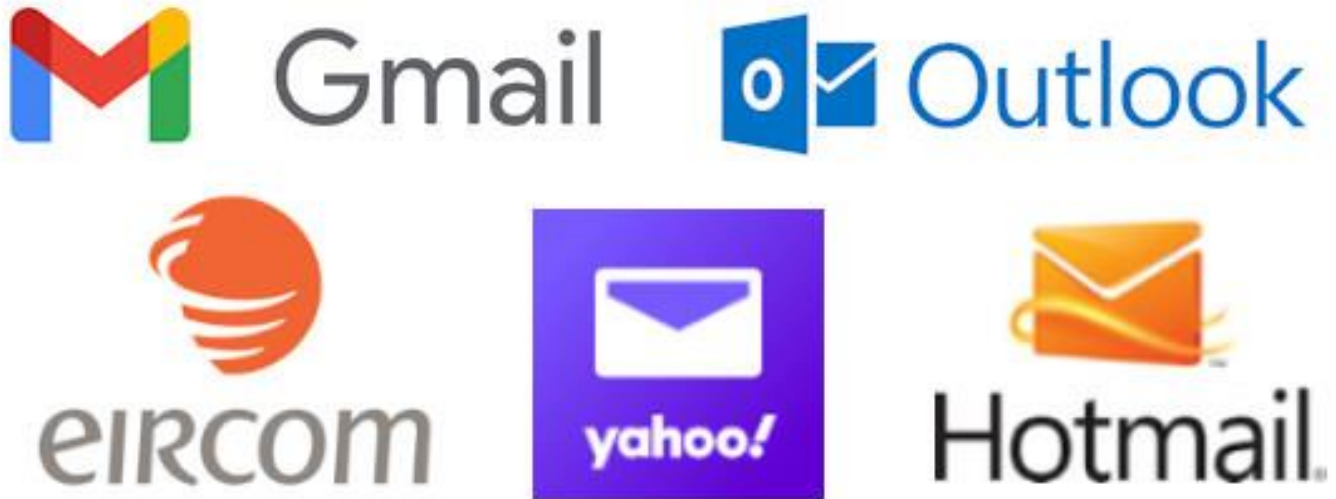
Nalanda Mahila College, Bihar Sharif

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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E-Mail



❖ What is Email?

Email (short for electronic mail) is a digital method of sending and receiving messages over the internet or other computer networks. It allows people to exchange information—such as text, files, images, and links—almost instantly, regardless of distance. Email has become one of the most widely used communication tools in the world for both personal and professional purposes.

➤ Email Address Structure

An email address has three main parts:

1. **Username:** The unique name chosen by the user (e.g., john.doe).
2. **@ symbol:** Separates the username from the domain.
3. **Domain name:** The server hosting the email service (e.g., gmail.com).

Syntax :- username@domain

Example: hirahiolal@gmail.com, hire@bharatpayaeps.online

Note :- email is not case-sensitive.

❖ **Email Domain Hierarchy**

1. TLD (Top-Level Domain)

The **last part** of a domain name.

Examples: .com, .org, .net, .in

2. Second-Level Domain (SLD)

The part **immediately before the TLD** in a simple domain.

Example: example.com → example is the SLD.

3. Third-Level Domain

An additional level between the SLD and the TLD in certain domain structures.

Example: example.co.in → co is a third-level domain label.

4. Subdomain

A domain created under an existing domain, usually placed before the main domain.

Example: mail.example.com → mail is a subdomain.

➤ **How Email Works**

Email operates using a client-server model. Email communication mainly uses **SMTP, DNS, and POP3/IMAP**. When you send an email:

1. Composing the Message

You create the email in an email application (such as Gmail, Outlook, Yahoo Mail) or on a web browser. You type the recipient's email address, subject, and message content/files.

2. Sending the Message

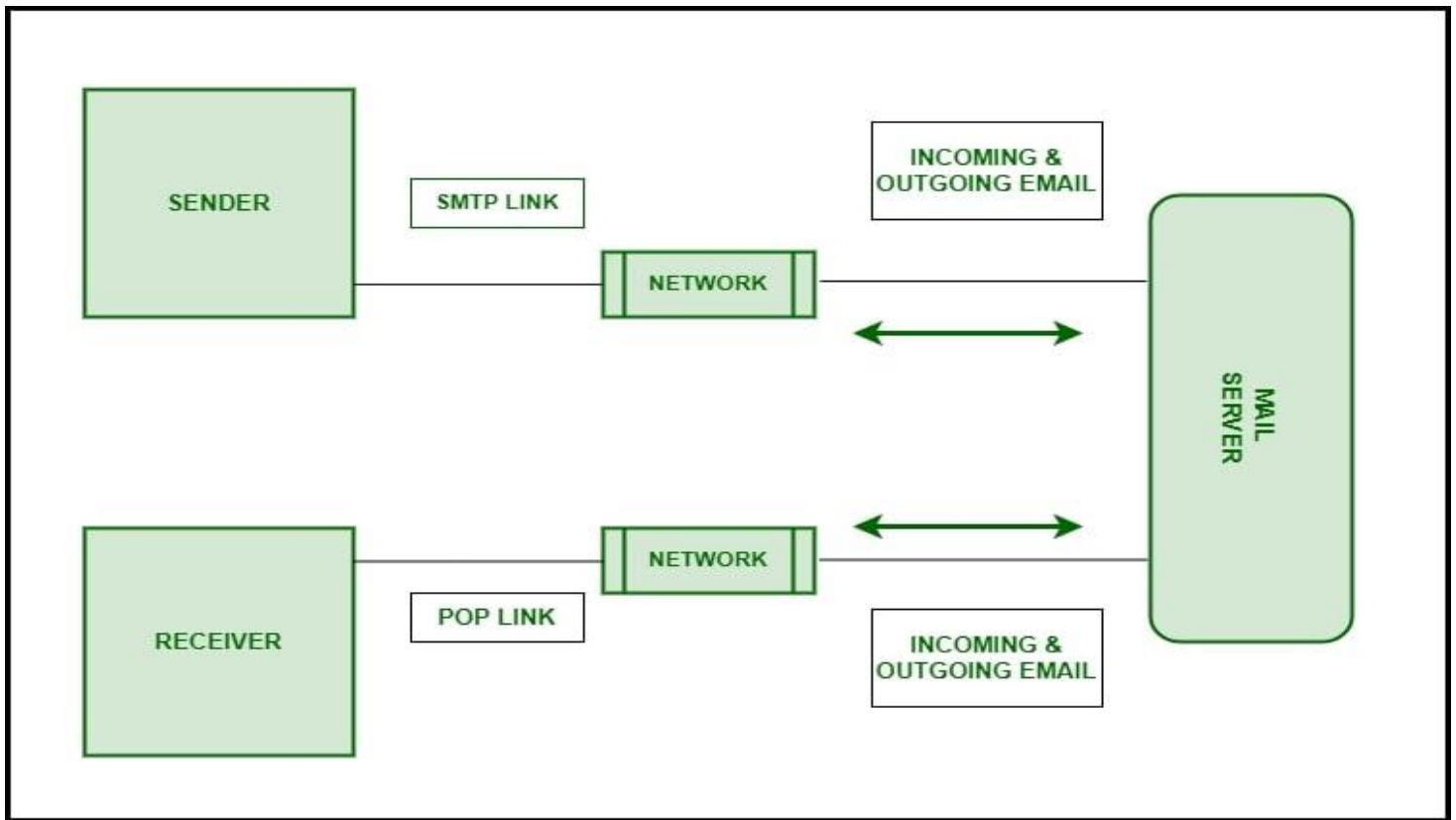
The email is sent from your device to an SMTP server (Simple Mail Transfer Protocol), which forwards it to the recipient's mail server.

3. Receiving the Message

The recipient's server stores the message until they check their inbox. They retrieve it using protocols like IMAP (Internet Message Access

Protocol) or POP3 (Post Office Protocol).

❖ Email working diagram



Protocol of Email = 3 Types

1. SMTP (Simple mail Transfer protocol)

- Used to send and transfer e-mails.
- Used when sending an email from the sender's mail client to the mail server.
- Also used for transferring email between mail servers.
- Example: Sending an email from Gmail to Yahoo.

2. IMAP (Internet message access protocol)

- Used to access and manage e-mails stored on the mail server.
- Emails are primarily stored on the mail server.
- It synchronizes email status across multiple devices.
- You can access the same mailbox from multiple devices.
- Actions such as read, unread, delete, move, etc. can be synchronized with the server.
- Suitable when you want your mailbox to remain synchronized across

devices.

- Example: Gmail, Yahoo Mail, Outlook can provide IMAP access.

(IMAP में email server पर रहती है और अलग-अलग devices पर mailbox को synchronize करके access किया जा सकता है।)

3. POP3 (Post office Protocol version 3)

→ POP3 is software based mail services. To retrieve / Receive email, for specific device not access any where, install software in this. This type of protocol all message save in server but one time seen any device then automatically message delete in server and permanent save in this device.

E.g – Apple, Microsoft,

MIME (Multipurpose internet mail)

→ It is not any email protocol. It is standard of email content/format attachments to represent. Used to support different types of content in email messages. Used for e-mail attachment files (.jpg, .mp3, .mp4, .pdf, .docx, Other file types)

❖ Types of Email Domain (based on their purpose or ownership)

1. Public Email Domain

A public email domain is an email domain that provides email services to the general public, usually free of cost.

e.g :- hirahiolal@gmail.com, apkaname@yahoo.com

Free Email ka domain

Sl.No	Email domain	Owner/Company /{storage}
1.	Gmail.com	Google {15GB}
2.	Yahoo.com	Yahoo! (now under Verizon) {1 TB}
3.	Hotmail.com /outlook.com	Microsoft {15GB}
4.	Aol.com	AOL (part of yahoo! /Verizon) {1TB*}
5.	Gmx.com / gmx.net	GMX internet services {65GB}

6.	Protonmail.com	Proton AG (Switzerland) {1 GB}
7.	Tutanota.com	Tutao Gmbh (Germany)
8.	Icloud.com	Apple inc. {5 GB}
9.	Zoho.com	Zoho corporation
10.	Yandex.com	Yandex
11.	Mail.com	1 & 1 IONOS
12.	Rediffmail.com	

Top public email list – gmail.com (17.74%) , yahoo.com(17.34%), Hotmail.com (15.53%), aol.com (3.2%)

2. Private / Custom Domains

A **private or custom email domain** uses a domain name owned by an individual, organization, or business for its email addresses. Companies, organisations, school use for personal branding.

e.g – support@amazon.com, ppup@helpenable.com, hire@bharatpayaeps.online,.....

3. Government domains

A **government email domain** is an email domain used by government departments and officials for official communication.

e.g – secretary@mea.gov.in,

4. Educational Domains

An **educational email domain** is a domain used by educational institutions such as schools, colleges, and universities for official communication.

e.g – hira@ppu.ac.in, student@gdc.edu,....

5. Country – code domains

A **country-code domain** is a domain extension assigned to a specific country or territory. Create for country specific TLD (Top level domain).

e.g – abc@firm.in,.....info@business.com.au (Australia),....

6. Temporary / Disposable Email

A temporary or disposable email is a short-term email address used for a limited time, usually to avoid spam or protect privacy. Mostly used in spam or testing.

e.g – temp@10minteemail.com, fake@mailinator.com,...



Choise Guide

Si.No	Used	Choose email Domain
1.	ज्यादा स्टोरेज के लिए	Yahoo / AOL
2.	Productivity & tools	Gmail / outlook
3.	Apple user	Icloud mail
4.	Business / Custom domain	Zoho mail
5.	Prefer security & privacy	Protonmail, tutanota, mailfence



Advantages of Email

1. Speed – Messages are delivered in seconds.
2. Cost-effective – Most email services are free.
3. Attachment support – Documents, images, videos, and other files can be sent easily.
4. Accessibility – Can be accessed from computers, smartphones, or tablets anywhere with internet access.
5. Record keeping – Emails can be stored, searched, and retrieved as needed.



Uses of Email

1. **Personal Communication:** Staying in touch with family and friends.
2. **Business Communication:** Sending reports, proposals, or meeting invites.
3. **Marketing:** Businesses send newsletters, offers, and promotions to customers.
4. **Official Documentation:** Sharing legal, academic, or corporate information.



Web – Based Email

- Those email which access in any browser (chrome, firefox, opera,.....) by using login id and password. It does not require a separate email application to access messages. Users can log in to

their email account from any device, such as a computer, laptop, tablet, or smartphone, using a web browser.

- Webmail stores email messages on the **mail server**, allowing users to access their emails from different devices. Webmail generally uses protocols such as **HTTP/HTTPS** for communication with the web interface and SMTP/IMAP for email services.
- E.g – Gmail, Yahoo, outlook, zoho, protonmail,

➤ **Limitations of Email**

1. **Spam and Junk Mail** – Unwanted advertising messages.
2. **Phishing Attacks** – Fraudulent emails trying to steal personal data.
3. **Overload** – Too many emails can overwhelm inboxes.
4. **Requires Internet** – Cannot be used offline except for drafting.

➤ **Security in Email**

To protect email communication, encryption methods such as SSL/TLS are used. Users are advised to use strong passwords, enable two-factor authentication (2FA), and be cautious when opening suspicious links or attachments. Regularly updating email software and avoiding public Wi-Fi networks can also help prevent unauthorized access and protect sensitive information.

❖ **Common Email Terms**

1. **CC (Carbon Copy)**

Used to send a copy of an email to other recipients. All recipients can see the CC addresses.

Example: Send an email to the Principal and CC the HOD.

2. **BCC (Blind Carbon Copy)**

Used to send a copy without showing the BCC recipients to others.

Example: Send an email to 50 students using BCC so their email addresses remain private.

3. **Subject**

A short title that tells the recipient what the email is about.

Example: Subject: BCA Practical Exam Notice

4. **Attachment**

A file sent along with an email.

Example: Attach a PDF, DOCX, or image to an email.

5. **Inbox**

The folder where **received emails** are stored.

Example: A student's reply to your email appears in the Inbox.

6. **Sent**

The folder where **emails you have sent** are stored.

Example: After sending a notice, you can find it in Sent.

7. **Draft**

An email that has been written but **not yet sent**.

Example: You write an email and close it without sending → it is saved as a Draft.

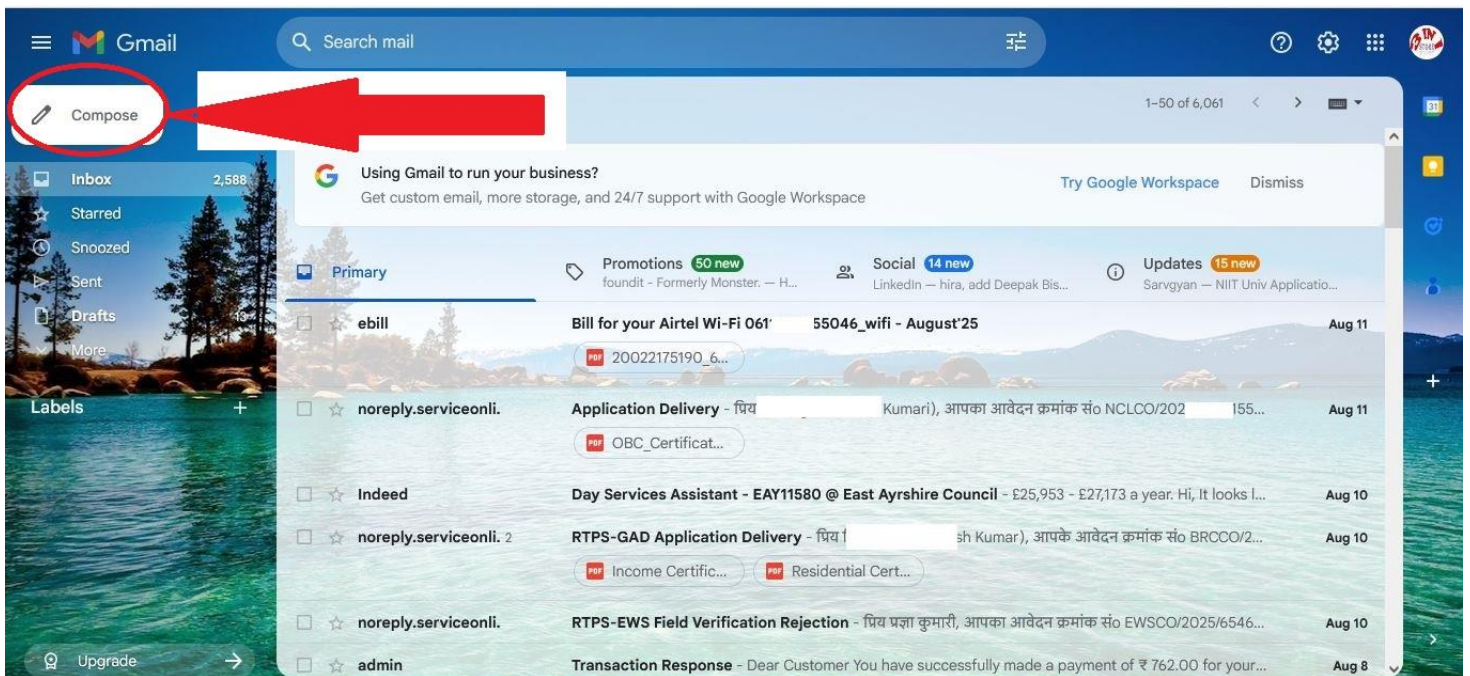
8. **Spam**

The folder where emails suspected to be **unwanted or suspicious** are placed.

Example: An unwanted promotional email may be moved to Spam.

➤ **How to Send an E-mail in Gmail to Multiple Recipients**

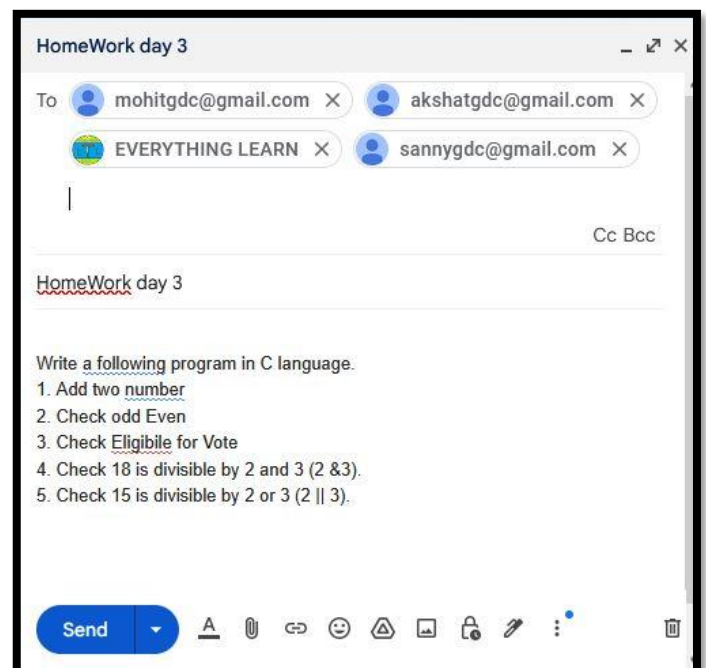
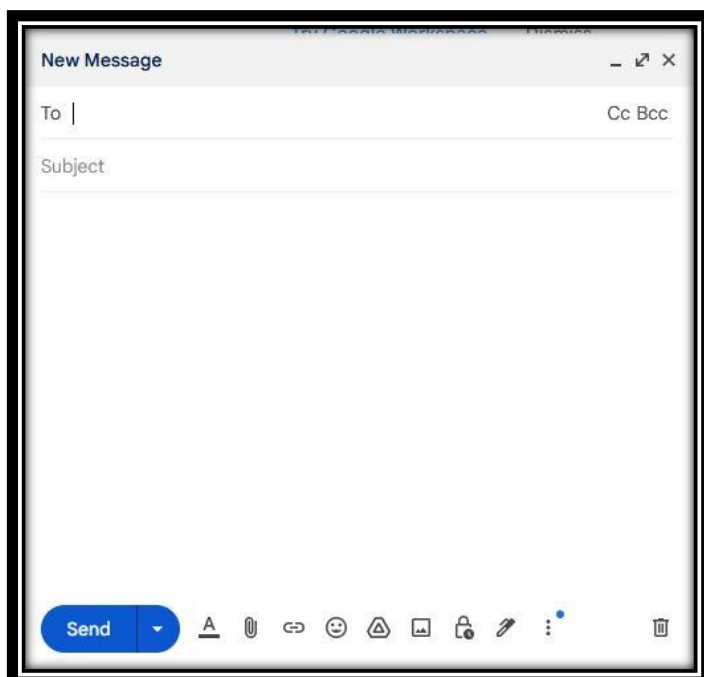
1. Click on Compose button. It follows the screen as shown in Fig.



Note :- E-mail messages are similar to letters, with two main parts:- 1. Header 2. Body

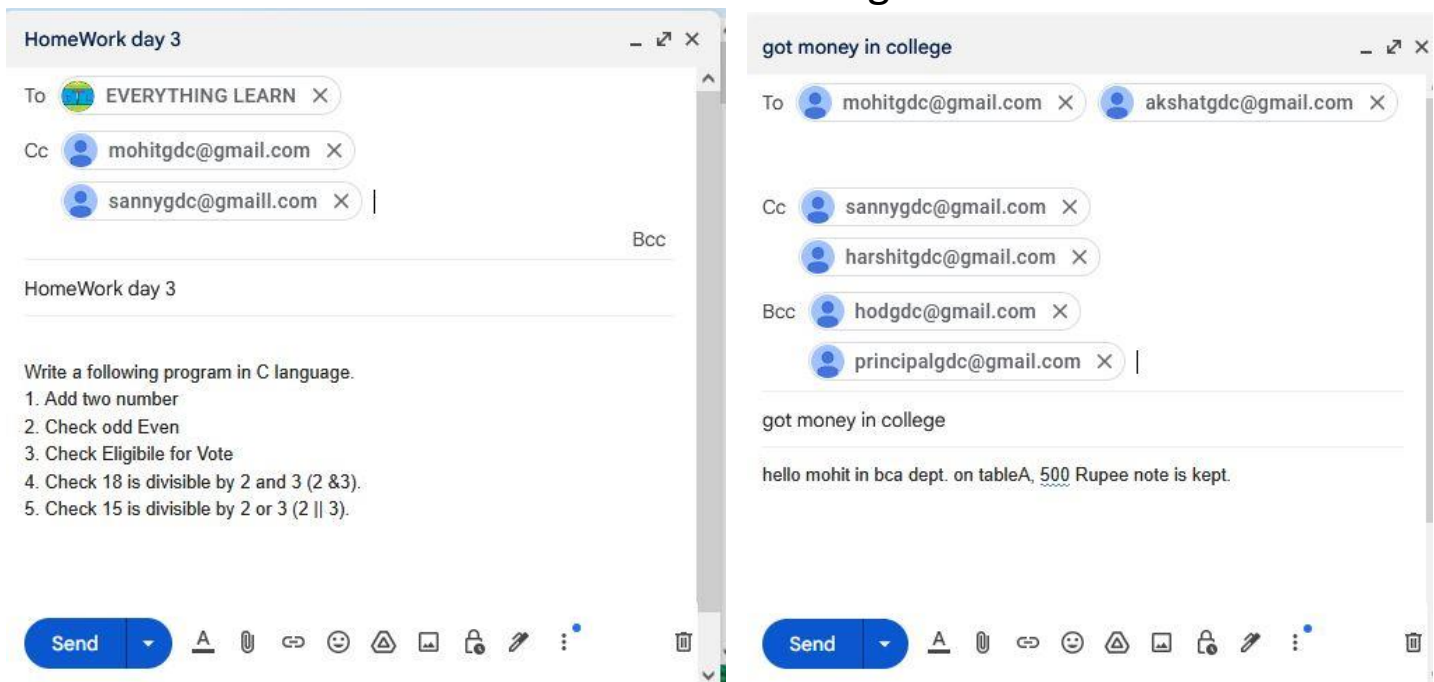
1. Header :- The header contains the name and address of the recipient, the address of anyone who is being copied, and the subject of the message.
2. Body :- The body contains the message itself. In header section

Type the e-mail address of the recipient in the “To:” field. For more than one person, e-mail id should separate by commas. At least one



3. Type the recipient's address in the 'To' field.

4. Type subject of your message in the “Subject:” field. Use the large text box to enter the contents of your message.
5. In the “cc:” field, enter the email address of those to whom you would like to send a “**carbon copy**” of your message. All other recipients of that message will be able to see that the person you designed as a “cc:” has received a copy of the message.
6. In the “Bcc:” field, enter the email address of those to whom you would like to send a “**blind carbon copy**” of your message. This is almost similar to the “Cc:” feature, except that “Bcc:” recipients are invisible to the “To:” and “Cc:” recipients of the message as well as to each other.
7. Click on Attach files for attaching file(s) and Insert Photos to include Photos if required.
8. Click on send button to send the message.



(b) What is E-mail ? Write steps to send a mail to many recipients.

(2021)

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(2023)

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